

ECO387C - Fall 2024

Discussion 1

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1 Partial Equilibrium

1.1 Setup

- Time is discrete and begins at $t = 1$ and goes until $t = 2$.
- There is a household (HH) that lives for two periods.
- There is only a single good that the household can consume or save.
- In each period, the household obtains an endowment (y_1, y_2) .
- In period 1, we allow households to trade a *nominal* bond (s) which yields $(1 + i)$ dollar in period 2.

The household's objective is to

$$\max_{c_1, c_2} U(c_1, c_2) = \max_{c_1, c_2} \ln(c_1) + \beta \ln(c_2) \quad (1)$$

subject to the budget constraints

$$p_1 c_1 + s = p_1 y_1 \quad (2)$$

$$p_2 c_2 = p_2 y_2 + (1 + i)s \quad (3)$$

Remarks on the setup:

- Note that the consumption good (c) and the endowment (y) share the same price. This is because they are the *same* good. However, bond (means of saving) has a separate market with a different unit so it needs a price of its own (interest rate).
- The price is set for each period. So even though the consumption and endowments are the same good, the price could vary across periods.

What we are concerned about is the optimal choice c_1 , c_2 , and saving s , taking prices p_1, p_2, i and endowments (y_1, y_2) as given. In this case, what matters is the *relative* price, not the absolute prices. Thus, we can also work with the budget constraint normalized by the price of consumption in each period:

$$p_1 c_1 + s = p_1 y_1 \Rightarrow c_1 + \frac{s}{p_1} = y_1 \Rightarrow c_1 + b = y_1 \quad (4)$$

$$p_2 c_2 = p_2 y_2 + (1+i)s \Rightarrow \frac{p_2}{p_2} c_2 = \frac{p_2}{p_2} y_2 + (1+i) \frac{p_1}{p_2} \frac{b}{p_1} \Rightarrow c_2 = y_2 + (1+r)b \quad (5)$$

The merit of normalization is that we only need to work with one price instead of three. Also, this is the *real* budget constraint where the LHS and the RHS are represented in units of consumption. b represents the *real* bond and r is real interest rate. The HH purchases the bond in period 1 and gets $1+r$ units of consumption in the second period (not in dollars).

1.2 Solution

Solving the optimization problem above means that we want to express the unknown choice variables (c_1, c_2, b) as a function of the knowns that agents take as given. In this case, these are price (r) and endowments y_1, y_2 . For the ease of notation, I will define bond price $q \equiv (1+r)^{-1}$. Then, we can combine the two budget constraints and set up the Lagrangian:

$$\max_{c_1, c_2} \ln(c_1) + \beta \ln(c_2) + \lambda (y_1 - c_1 - q(c_2 - y_2)) \quad (6)$$

Taking the first-order conditions:

$$c_1 : \frac{1}{c_1} = \lambda \quad (7)$$

$$c_2 : \frac{\beta}{c_2} = \lambda q \quad (8)$$

$$\lambda : c_1 + q(c_2 - y_2) = y_1 \quad (9)$$

Combining the first two leads to the celebrated Euler equation:

$$\frac{1}{c_1} = \beta \frac{1}{c_2} q^{-1} \Leftrightarrow \frac{1}{c_1} = \beta \frac{1}{c_2} (1+r) \quad (10)$$

Using the budget constraint gives us the following solution:

$$\begin{cases} c_1^* = \frac{y_1 + q y_2}{1+\beta} \\ c_2^* = \frac{\beta(y_1 + q y_2)}{q(1+\beta)} \\ b^* = \frac{\beta y_1 - q y_2}{1+\beta} \end{cases} \quad (11)$$

These are the optimal consumption-saving choices of the HH, taking price and endowments as given.

2 General Equilibrium

In a partial equilibrium setup, the focus is on the optimal behavior of a single agent (household). In contrast, general equilibrium (or simply 'equilibrium') involves aggregating the optimal decisions of multiple individuals while ensuring that aggregate consumption is consistent with the aggregate endowment. To illustrate, I will discuss a simple example with two agents, where the only price to consider is the price of the real bond or real interest rate, denoted by $q = (1 + r)^{-1}$.

2.1 Setup and Definition

Suppose there are two agents, A and B , who face identical optimization problem (1), subject to the normalized, period-by-period real constraints, (4) and (5). The only difference between the two agents is the stream of endowments. The endowments for agents A and B are denoted by (y_1^A, y_2^A) and (y_1^B, y_2^B) , respectively, where the subscripts denote time and the superscripts denote agents. Consequently, the aggregate endowments in each period are $Y_1 = y_1^A + y_1^B$ and $Y_2 = y_2^A + y_2^B$.

We can then ask how to allocate the aggregate endowment in each period to each agent (A , B) in a way that satisfies both agents (optimizing their consumption) without leaving any endowment unallocated. Individuals take prices as given and determine their optimal level of consumption. However, the sum of consumption may exceed or fall short of the aggregate endowment, even if individually optimal. The role of equilibrium is to find the price that aligns total consumption with the aggregate endowment, which is known as the market clearing condition. Thus, equilibrium consists of two conditions: individual optimality and market clearing.

Definition 1 (Competitive Equilibrium) *A competitive equilibrium is a set of price $\{q\}$ and a set of allocations $\{c_1^A, c_1^B, c_2^A, c_2^B, b^A, b^B\}$ such that*

1. *Individual optimality: Taking price q as given, the allocations $\{c_1^A, c_1^B, c_2^A, c_2^B, b^A, b^B\}$ solve the household's problem*
2. *Market clearing: The market for goods clears*

$$c_t^A + c_t^B = y_t^A + y_t^B \quad \text{for each } t = 1, 2$$

and the market for assets (bonds) clears

$$b^A + b^B = 0$$

2.2 Individual Optimality

First, note that both agents face the same problem except for the differences in the streams of endowments. This means the optimal consumption choice of each agent is

$$\text{Household A} \begin{cases} c_1^A = \frac{y_1^A + qy_2^A}{1+\beta} \\ c_2^A = \frac{\beta(y_1^A + qy_2^A)}{q(1+\beta)} \\ b^A = \frac{\beta y_1^A - qy_2^A}{1+\beta} \end{cases} \quad \text{Household B} \begin{cases} c_1^B = \frac{y_1^B + qy_2^B}{1+\beta} \\ c_2^B = \frac{\beta(y_1^B + qy_2^B)}{q(1+\beta)} \\ b^B = \frac{\beta y_1^B - qy_2^B}{1+\beta} \end{cases}$$

Notice that each agent's optimal consumption and saving decisions depend on the price (q) and their respective individual endowment (y). As long as the consumption-saving decisions are consistent with these, the first condition of equilibrium is satisfied.

2.3 Market Clearing

The optimal consumption and saving functions outlined above suggest that households make their choices by taking price and their individual endowments as given. However, in equilibrium, we want to allocate the entire aggregate endowment to each household for consumption. In economic terms, this requires 'market clearing', ensuring that aggregate demand (e.g. the sum of consumption) aligns with aggregate supply (e.g. the sum of endowment). There are two markets to consider: Goods and Bonds. The market-clearing conditions for each market are:

$$\begin{cases} \text{Goods in period 1:} & c_1^A + c_1^B = Y_1 \equiv y_1^A + y_1^B \\ \text{Goods in period 2:} & c_2^A + c_2^B = Y_2 \equiv y_2^A + y_2^B \\ \text{Bond:} & b^A + b^B = 0 \end{cases}$$

By plugging the optimal consumption and saving functions into the market-clearing conditions, we derive the equilibrium price:

$$q = \beta \frac{Y_1}{Y_2} \equiv \beta \frac{y_1^A + y_1^B}{y_2^A + y_2^B}$$

The equilibrium price (inverse of real interest rate), depends on the aggregate endowment. If the aggregate endowment is fixed over time ($Y_1 = Y_2$), the real interest rate will be equal to β .

2.4 Examples

Consider the following numerical examples:

1. $(y_1^A, y_2^A) = (1, 2)$ and $(y_1^B, y_2^B) = (2, 1)$: There is no aggregate fluctuation ($Y_1 = Y_2 = 3$). The equilibrium price and associated consumption-saving allocations are

$$\begin{cases} \text{Price: } q = \beta \\ \text{Period 1 consumption: } c_1^A = \frac{1+2\beta}{1+\beta}, & c_1^B = \frac{2+\beta}{1+\beta} \\ \text{Period 2 consumption: } c_2^A = \frac{1+2\beta}{1+\beta}, & c_2^B = \frac{2+\beta}{1+\beta} \\ \text{Bond (Saving): } b^A = \frac{-\beta}{(1+\beta)} & b^B = \frac{\beta}{(1+\beta)} \end{cases}$$

In period 1, the poorer HH A borrows from the richer HH B. Both households achieve the same level of consumption in both periods (consumption-smoothing).

2. $(y_1^A, y_2^A) = (2, 1)$ and $(y_1^B, y_2^B) = (1, 3)$: There is aggregate fluctuation ($Y_1 = 3 < Y_2 = 4$). The equilibrium price and associated consumption-saving allocations are

$$\begin{cases} \text{Price: } q = \frac{3\beta}{4} \\ \text{Period 1 consumption: } c_1^A = \frac{8+3\beta}{4(1+\beta)}, & c_1^B = \frac{4+9\beta}{4(1+\beta)} \\ \text{Period 2 consumption: } c_2^A = \frac{8+3\beta}{3(1+\beta)}, & c_2^B = \frac{4+9\beta}{3(1+\beta)} \\ \text{Bond (Saving): } b^A = \frac{5}{4(1+\beta)} & b^B = \frac{-5}{4(1+\beta)} \end{cases}$$

In period 1, the poorer HH B borrows from the richer HH A. Notice that both households achieve higher consumption in period 2, when the aggregate endowment is higher.

3. $(y_1^A, y_2^A) = (1, 1)$ and $(y_1^B, y_2^B) = (1, 1)$: There is no aggregate fluctuation ($Y_1 = Y_2 = 2$). The equilibrium price and associated consumption-saving allocations are

$$\begin{cases} \text{Price: } q = \beta \\ \text{Period 1 consumption: } c_1^A = 1, & c_1^B = 1 \\ \text{Period 2 consumption: } c_2^A = 1, & c_2^B = 1 \\ \text{Bond (Saving): } b^A = 0 & b^B = 0 \end{cases}$$

There is no heterogeneity between A and B. So there is no borrowing or lending.

4. $(y_1^A, y_2^A) = (0, 0)$ and $(y_1^B, y_2^B) = (2, 2)$: There is no aggregate fluctuation ($Y_1 = Y_2 = 2$). The equilibrium price and associated consumption-saving allocations are

$$\begin{cases} \text{Price: } q = \beta \\ \text{Period 1 consumption: } c_1^A = 0, & c_1^B = 2 \\ \text{Period 2 consumption: } c_2^A = 0, & c_2^B = 2 \\ \text{Bond (Saving): } b^A = 0 & b^B = 0 \end{cases}$$

There is heterogeneity between A and B. But B is always richer than A. For B, there is nothing to gain from trading with A. So B just consumes his own endowment in each period.